Enhancing Distribution System State Estimation Under Limited Measurements: Leveraging Large Language Model and Multimodal Information

Mingyang Gao, *Student Member*, *IEEE*, Suyang Zhou, *Member*, *IEEE*, Wei Gu *Senior Member*, *IEEE*, Jili Fan, Aobo Guan, *Student Member*, *IEEE*, Hong Zhu, Lei Wei, and Zijian Hu

***Abstract*—Accurate distribution system state estimation (DSSE) under limited measurement scenarios remains a critical challenge due to the inherent high-dimensional nonlinearity and data scarcity in power distribution networks. Leveraging the exceptional few-shot pattern recognition capabilities of pre-trained large language models (LLMs), this paper proposes a novel DSSE framework that synergizes LLMs with multimodal data integration, enabling effective voltage magnitude and phase angle estimation. Moreover, this study introduces a prompt engineering approach that transforms statistical features extracted from historical data into contextual prompts. By employing natural language representations, this method enhances temporal feature extraction and uncovers latent patterns in scenarios with limited measurements. Furthermore, a channel-independent multimodal fusion architecture that structurally aligns time-series measurements, real-time sensor data, and textual prompts while preserving modality-specific characteristics. Extensive validation on IEEE 33-bus and Simbench 144-bus systems demonstrates the effectiveness of the proposed method, which reduces MAE by 17.7%–25.1% and RMSE by 17.3%–18.2% compared to the second best data-driven baselines under limited measurement scenarios. These results highlight the framework's strong generalizability across diverse network topologies and its potential for practical deployment in poorly instrumented distribution networks. Our codes are available at https://github.com/gmy1997ee/LLM-MultiModal-for-DSSE.**

***Index Terms*—Distribution system state estimation, large language model, limited measurements, multimodal data.**

I. Introduction

The distribution system plays a vital role as a fundamental component in modern power systems, serving as the backbone for electrical energy delivery and transmission [1]. The implementation of accurate and reliable distribution system state estimation (DSSE) is paramount to maintain system stability and operational efficiency [2]. Distribution networks pose unique challenges relative to transmission systems, including complex topologies, sparse measurement infrastructure, and heterogeneous data quality [3]. In particular, constraints such as the high cost of installation and maintenance often result in the "sparse measurement" problem, which challenges the accuracy and reliability of state estimation methods traditionally employed in the power system [4]. Furthermore, the random and fluctuating nature of load characteristics in distribution systems, combined with the massive integration of modern electrical devices such as distributed energy resources and electric vehicles, exacerbates the difficulty of state estimation [5]. Consequently, improving the accuracy of state estimation under sparse measurement scenarios has become a critical scientific problem that demands immediate resolution.

Traditional methodologies for DSSE rely on high-quality monitoring data and mathematical optimization models [6]. The Weighted Least Squares (WLS) based approach has been widely regarded as a fundamental approach to addressing this challenge [7]. This method achieves estimation by solving the state estimation equations to convergence using measurement data from metered nodes and pseudo-measurement data (obtained through prediction or other methods) for unmetered nodes. However, non-Gaussian noise or missing data can prevent WLS convergence, rendering accurate state estimation unattainable. Moreover, since the pseudo-measurement data used in the state estimation process is derived from predictive models, the accuracy of the DSSE results is inherently dependent on the errors of the predictive models [8].

In recent years, with the rapid development of artificial intelligence technologies, state estimation methods based on deep learning have demonstrated higher accuracy and faster convergence rates [9], [10]. These methods are capable of extracting data features and establishing nonlinear mapping relationships, exhibiting superior performance when dealing with complex systems [11]. Building on this foundation, a variety of neural network architectures have been introduced to address the state estimation task, tailored to the diverse data structures within state estimation systems. In [12], the authors proposed a framework utilizing fully connected neural networks for state estimation and validated the superiority of data-driven approaches over WLS methods in under-measured scenarios. To better leverage the time-series data in DSSE scenarios, [13] employed LSTM networks, commonly used

Manuscript received March 25, 2025; revised May 24, 2025; accepted June 24, 2025. Date of online publication November 5, 2025; date of current version January 18, 2026. This work is supported by the National Key Research and Development Program (2022YFB2404200).

M. Y. Gao, S. Y. Zhou (corresponding author, email: suyang.zhou@seu.edu.cn), W. Gu, J. L. Fan and A. B. Guan are with the School of Electrical Engineering, Southeast University, Nanjing 210096, China.

Z. J. Hu and H. Zhu are with Nanjing Power Supply Company, State Grid Jiangsu Electric Power Co., Ltd., Nanjing 210019, China.

L. Wei is with State Grid Jiangsu Electric Power Co., Ltd., Nanjing 210019, China.

DOI: 10.17775/CSEEJPES.2025.01460

2096-0042 © 2025 CSEE. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

in time-series forecasting, to perform state estimation. This approach effectively extracted latent state variable distribution features from historical operational data, mitigating the challenges posed by insufficient real-time measurements.

Explicit physical information in power grids can also enhance the accuracy of state estimation in under-measured scenarios. [14] demonstrated that, given known distribution network topology, using Graph Neural Networks (GNNs) for DSSE achieves higher estimation accuracy compared to MLPs. Similarly, [15] employed GNNs for state estimation and showed that this architecture performs better in noisy environments and scenarios with missing measurements. In [16], the authors not only incorporated topological information but also introduced a power flow constraint loss function into the state estimation task. They utilized a physics-informed graph neural network (PIGNN) to complete the state estimation. Furthermore, [17] replaced power flow equations with GNNs to directly model the nonlinear mapping between measurements and estimated states. By constructing a weakly supervised learning framework and a loss function based on the WLS process, the network's estimation accuracy was further improved. The integration of time-series information, topological information, and other physical data has significantly enhanced the performance of state estimation tasks [18], [19]. However, existing deep learning methods often require large amounts of training data to ensure the accuracy of DSSE tasks.

Recently, the emergence of ChatGPT has garnered widespread attention, as LLMs have demonstrated exceptional accuracy in various sequence generation tasks [20], [21]. Several research institutions have open-sourced pre-trained large models trained on billions of data points [22], validating their fundamental causal reasoning capabilities [23]. Studies [24], [25] provide detailed experimental results, demonstrating the superiority of LLMs over conventional deep learning methods in time series prediction tasks, particularly in zero-shot and few-shot scenarios. These results underscore the robust data extraction and reasoning capacity of LLMs under limited-data scenarios. While [26] demonstrated the effectiveness of LLMs in DSSE through a multi-task learning framework, their study did not investigate the estimation capabilities of LLMs in limited-measurement scenarios, nor did it address the processing of multimodal data. Table I provides a comparative summary of existing data-driven DSSE methods and the approach proposed in this paper, with particular emphasis on the types of data utilized.

Inspired by above studies, this paper proposes a framework for employing LLMs in state estimation under scenarios with limited measurements. It proposes a novel paradigm for incorporating multimodal data into DSSE tasks, presenting a highly generalizable DSSE approach. The main contributions of this paper are summarized as follows:

1) A novel state estimation methodology for distribution systems leveraging fine-tuned large language models is proposed. The approach capitalizes on the pre-trained knowledge of LLMs to achieve accurate state estimation under limited measurement conditions.

2) A specialized text prompt modeling approach for state estimation tasks is proposed, which transforms multi-time-scale statistical features from historical operational data into contextualized prompts. This novel representation enables enhanced information extraction from limited datasets, effectively achieving feature augmentation through natural language encoding.

3) A channel-independent multimodal fusion framework is proposed for structured integration of time-series data, real-time measurements, and textual prompts in DSSE tasks. The architecture preserves modality-specific characteristics while enhancing state estimation accuracy without compromising computational efficiency.

4) The accuracy of the proposed DSSE method is validated on two open-source distribution systems of different scales, including the IEEE 33-bus system and the 144-bus network from Simbench [27]. Notably, under identical input conditions, the model achieves the best estimation accuracy across MAE and RMSE metrics.

II. Problem Formulation

A. Model-based DSSE Approach

The conventional model-based state estimation process can be formulated as an optimization problem. Based on power system flow equations, the relationship between state variables v and measurement values z can be expressed as:

$$z = \boldsymbol{h}(v) + e \tag{1}$$

where e represents measurement errors.

The state estimation solution v can be obtained by solving the following optimization problem:

$$\min_{v} J(v) = \sum_{i=1}^{m} w_i (z_i - \boldsymbol{h}_i(v))^2 \tag{2}$$

where $w_i = 1/\sigma_i^2$ is the weight of the i-th measurement and $\boldsymbol{h}(\cdot)$ is the measurement function.

TABLE I
DATA-DRIVEN DSSE METHODS WITH ITS DATA IMPLEMENTS

Methods	Real time measurement data	Historical time series data	Topological information	Physical constraints	Textual prompts
MLP [12]	●	○	○	○	○
LSTM [13]	○	●	○	○	○
GNN [14]	●	○	●	○	○
PIGNN [16]	●	○	●	●	○
LLMSE [26]	○	●	○	○	●
Ours	●	●	●	●	●

● denotes that the data type is utilized by the method; ○ denotes that it is not utilized.

$$\boldsymbol{h}(v)=\begin{cases} P_i = V_i \sum_{j=1}^{n} V_j(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}) \\ Q_i = V_i \sum_{j=1}^{n} V_j(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}) \\ P_{ij} = V_i^2(g_{ij}+g_{si}) \\ \qquad - V_iV_j(g_{ij}\cos\theta_{ij} + b_{ij}\sin\theta_{ij}) \\ Q_{ij} = -V_i^2(b_{ij}+b_{si}) \\ \qquad - V_iV_j(g_{ij}\sin\theta_{ij} - b_{ij}\cos\theta_{ij}) \end{cases} \tag{3}$$

where P_i, Q_i are the active and reactive power injections at bus i; P_{ij}, Q_{ij} are the active and reactive power flows from bus i to bus j; V_i, V_j are the voltage magnitudes at bus i and bus j; θ_{ij} is the voltage angle difference between bus i and bus j ($\theta_i - \theta_j$); G_{ij}, B_{ij} are the real and imaginary parts of the (i,j)th element in the bus admittance matrix; g_{ij}, b_{ij} are the conductance and susceptance of the branch between bus i and bus j; g_{si}, b_{si} are the conductance and susceptance of the shunt element at bus i; n is the total number of buses in the power system.

B. Deep Learning-based DSSE Approaches

Equation 1 presents the fundamental solution model for state estimation tasks, where the function $\boldsymbol{h}(\cdot)$ represents the critical measurement model that establishes the relationship between measurements and estimated variables. State estimation methods based on deep learning employ a neural network with learnable parameters to construct this measurement function, specifically modeling the relationship between z and v. This process is implemented through a supervised learning framework, which can be formally expressed as:

$$\theta^* = \arg\min_{\theta} \sum_{i=1}^{m} (z_i - f(v_i))^2 \tag{4}$$

where $f(\cdot)$ is the neural network model, θ represents the model parameters, and z_i and v_i denote the i-th measurement and state variable, respectively.

In real-world power grid scenarios, real-time measurements are often insufficient, particularly in distribution network where measurement devices are scarce resources. As a result, improving state estimation accuracy with limited data has increasingly attracted researchers' attention. Considering the nature of data collected by sensors in practical scenarios, we primarily utilize two types of data to accomplish this task: historical operational data in time series format and real-time measurement data from nodes equipped with measurement devices within the network.

Furthermore, if the distribution network topology is known, this information can be incorporated into the state estimation process through graph neural networks. Based on the above considerations, the deep learning-based state estimation process utilizing multi-source data can be formally expressed as:

$$v_t = f_\theta(\{v_{t-k:t-1}\}, z_t, \mathcal{G}) \tag{5}$$

where v_t denotes the estimated state vector at time step t, $\{v_{t-k:t-1}\}$ represents the historical state sequence over the past k time steps, z_t represents the real-time measurement vector and $\mathcal{G}$ denotes the network topology of the system.

C. Advantages of Integrating LLMs for DSSE

The pre-trained parameters in LLMs are derived from vast contextual data, embedding the model's capability to reason based on logical and causal relationships between contexts. Recent studies have successfully applied LLMs and their variants to time series forecasting tasks, achieving state-of-the-art (SOTA) accuracy. While both time series prediction and state estimation fall under numerical regression tasks, they differ in that state estimation generates instantaneous values for system variables at a specific time step, rather than predicting future sequences. These studies reveal several advantages of utilizing LLMs for regression tasks:

1) Enhanced Data Mining Capability through Multimodal Data: LLMs possess an inherent ability to process multimodal data, simultaneously handling various input types such as numerical values and text. This enables feature extraction across multiple dimensions, thereby enhancing system state understanding and estimation accuracy.

2) Robust Few-shot Learning Capability: Pre-trained models incorporate extensive domain knowledge and universal patterns. Through fine-tuning, they can rapidly adapt to specific state estimation tasks, maintaining strong performance even with limited training data.

3) Resilience to Noise and Anomalies: During pre-training, LLMs are exposed to substantial noisy data, developing natural immunity to anomalies and noise. This characteristic proves particularly valuable in state estimation scenarios where noise is an inherent factor, resulting in enhanced robustness.

Inspired by these findings, this paper proposes a DSSE method that integrates large language models with multimodal data. Following the incorporation of LLMs, the state estimation model framework is redefined as follows:

$$v_t = f_{\theta_{\text{FT}},\theta_{\text{Pre-trained}}}(v_{t-k:t-1}, z_t, c_t, \mathcal{G}) \tag{6}$$

where c_t is the textual input providing contextual or prior information at time step t, θ_{FT} denotes the fine-tuned parameters, and θ_{Frozen} represents the frozen parameters of the pre-trained LLM.

III. Proposed Method

This section presents a detailed methodology for state estimation using LLMs. First, we introduce the specific architecture of our proposed large language model. Subsequently, we describe the construction method for text prompts applicable to state estimation tasks and the coupling strategy between multimodal input data and labels during supervised fine-tuning. Finally, we provide a comprehensive description of our proposed channel independent processing method for multimodal data.

A. Model Architecture

Considering the data available in real-world DSSE scenarios, this study utilizes three types of information as inputs for the DSSE model: textual data, historical operational time series data, and real-time measurement data from measured nodes. The i-th input data for DSSE can be represented as:

$$z_i = (c_i, t_i, m_i) \tag{7}$$

For the LLM, inspired by the findings in [28], we select the GPT-2 model as the foundation to process textual inputs [29]. The overall framework of the DSSE model is illustrated in Fig. 1. First, we use the pre-trained GPT-2 model to process the textual prompt inputs c_i, incorporating positional encoding for each token into the tokenized results, expressed as:

$$e_i = \text{WordEmbedding}(c_i) + \text{PositionalEmbedding}(i) \quad (8)$$

where $e_i \in \mathbb{R}^{L_p \times N_{\text{gpt2}}}$ represents the embedding vector of the i-th prompt, L_p denotes the length of the prompt, N_{gpt2} is the dimension of the GPT-2 embedding vector, which is set to 768 in our research, and c_i denotes the i-th prompt.

Subsequently, we employ the Transformer Layers of the GPT-2 model to process and map the input into high-dimensional semantic representations using pre-trained parameters. The Transformer Layers consist of multi-head attention modules, feed-forward modules, and layer normalization modules. In our experiment, we stack 12 Transformer Layers with pre-trained parameters, which can be expressed as:

$$h'_{p_i} = \text{LayerNorm}(\text{MultiHeadAttention}(e_i) + e_i) \quad (9)$$

$$h_{p_i} = \text{LayerNorm}(\text{FeedForward}(h'_{p_i}) + h'_{p_i}) \quad (10)$$

where $h'_{p_i} \in \mathbb{R}^{L_p \times N_{\text{gpt2}}}$ denotes the intermediate output and $h_{p_i} \in \mathbb{R}^{L_p \times N_{\text{gpt2}}}$ represents the output of Transformer Layer.

Concurrently, we construct a time series input layer and a real-time measurement input layer to interface with the corresponding time series data $t_i \in \mathbb{R}^{L_t \times N_v}$ and real-time measurement data $m_i \in \mathbb{R}^{L_d \times N_v}$. Each of these data types is mapped to the same dimensionality as h_p through a dedicated alignment layer, which can be expressed as:

$$h_{t_i} = \text{AlignmentLayer}(\text{TimeSeriesInputLayer}(t_i)) \quad (11)$$

$$h_{m_i} = \text{AlignmentLayer}(\text{RTMeasurementInputLayer}(m_i)) \quad (12)$$

where L_t is the length of the time series data, L_d is the length of the real-time measurement data, N_v is the number of variables, $h_{t_i} \in \mathbb{R}^{L_t \times N_{\text{gpt2}}}$ and $h_{m_i} \in \mathbb{R}^{L_d \times N_{\text{gpt2}}}$ are the high-dimensional representations of time series and real-time measurement data, respectively.

We combine the obtained hidden states h_{p_i}, h_{t_i}, and h_{m_i} to derive the fused feature $h_{f_i} \in \mathbb{R}^{(L_p+L_t+L_d) \times N_{\text{gpt2}}}$. Finally, h_f is mapped to the target shape of the state estimation results through an output layer, which can be expressed as:

$$h_{f_i} = \text{Concatenate}(h_{p_i}, h_{t_i}, h_{m_i}) \quad (13)$$

$$v_i = \text{OutputLayer}(h_{f_i}) \quad (14)$$

B. Multi-modal Data Integration for DSSE

In state estimation tasks based on LLMs, the construction of textual prompts significantly influences model performance. Well-designed prompts not only guide the model toward better input data comprehension but also implicitly integrate domain knowledge and physical constraints into the state estimation process. To provide LLMs with sufficient information while maximizing the distinctive features between different prompts, this paper constructs textual prompts using three aspects:

1) Time Series Input Description: Multiple experimental studies in time series prediction have demonstrated that textual descriptions of time series characteristics enhance prediction accuracy [30]. This paper implements differentiated textual descriptions of time series data, focusing on both short-term and long-term trends. Five key terms describe the sequence trends: *Sharp Increase, Slight Increase, Stable, Slight Decrease, and Sharp Decrease.*

For all time series, we determine long-term trends by analyzing the overall slope distribution, while short-term trends are assessed using the final 25% of the sequence. Fig. 1 visualizes different trend patterns, providing a more intuitive demonstration of the textual distinctions.

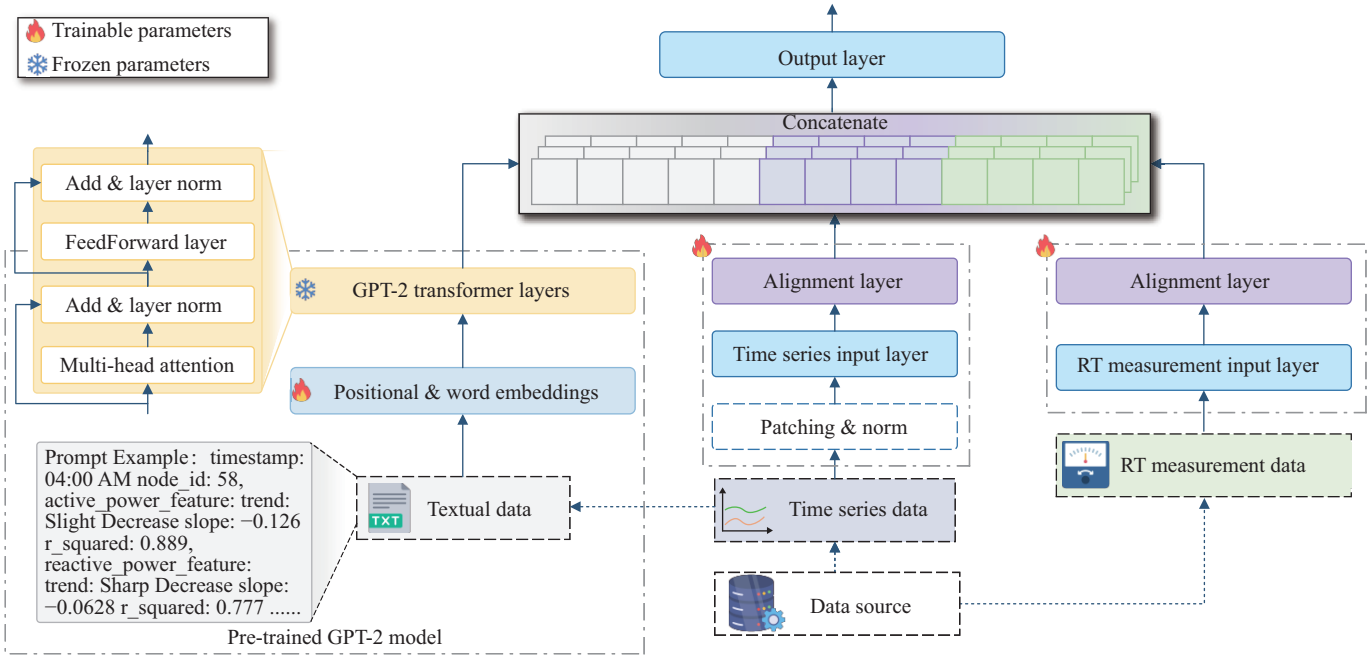


Fig. 1. The architecture of the proposed DSSE model based on GPT-2 and multimodal data.

2) Topological Information: We embed each node's topological relationships as textual prompts within the data. Specifically, the connections between each node and its neighboring nodes through specific edges are documented in detail.

3) Constraint Information: The textual format provides a more flexible approach to incorporating power grid constraints into model reasoning. For each node, voltage upper and lower limits can be directly input into the DSSE model through textual expressions

C. Channel-Independent Multimodal Data Processing Mechanism

In models that involve multimodal data inputs, cross-modal alignment mechanisms are critical to the overall performance of the model. In previous numerical regression tasks, such as state estimation [26] and time series forecasting [28] based on LLMs, numerical values are typically mapped to the same dimensionality as word embeddings (N_{gpt2}). These numerical representations are then concatenated with word embeddings and subsequently fed into the LLM.

However, since the pre-training data of LLMs primarily consists of textual data, recent studies have shown that directly embedding numerical information into LLMs often fails to improve task performance [31]. Additionally, because each input vector in an LLM is mapped to a 768-dimensional space, increasing the input length significantly amplifies computational costs. Unlike previous studies, this paper proposes and applies a channel-independent processing mechanism for multimodal data. Specifically, this approach ensures that the LLM processes only textual prompt inputs, thereby strictly preserving the effectiveness of the pretrained parameters in each layer of the GPT-2 model.

For numerical data, we construct distinct input layers tailored to different data types.

Specifically, for a time series input t_i of length L_t, the sequence is first divided into N patches, denoted as $p_i \in \mathbb{R}^{Lp\times N}$. Each patch p_i is then passed through a linear layer, where it is mapped to a higher-dimensional space and augmented with a positional encoding. This process is expressed as:

$$h_{p_i} = W_p p_i + W_{\text{pos}} \tag{15}$$

where $W_p \in \mathbb{R}^{N_{\text{gpt2}}\times L_p}$ is the weight matrix of the linear layer, $W_{\text{pos}} \in \mathbb{R}^{N_{\text{gpt2}}\times N}$ is the positional encoding matrix, and $h_{p_i} \in \mathbb{R}^{N_{\text{gpt2}}\times N}$ is the embedding vector of the i-th patch.

Next, a multi-head attention layer is applied to process h_{p_i}:

$$O_{p_i} = \text{Attention}(h_{p_i}) = \text{Softmax}\left(\frac{QK^{\text{T}}}{\sqrt{d_k}}\right)V \tag{16}$$

where $Q \in \mathbb{R}^{N_{\text{gpt2}}\times d_k}$, $K \in \mathbb{R}^{N_{\text{gpt2}}\times d_k}$, and $V \in \mathbb{R}^{N_{\text{gpt2}}\times d_v}$ are the query, key, and value matrices, respectively, d_k is the dimension of the key matrix, and $O_{p_i} \in \mathbb{R}^{N_{\text{gpt2}}\times d_v}$ is the output of the attention layer.

Finally, the output of the attention layer is passed through an alignment layer to generate the final embedding vector $h_{t_i} \in \mathbb{R}^{Lt\times N_{\text{gpt2}}}$:

$$h_{t_i} = (W_O(O_{p_i})^{\text{T}} + b) \tag{17}$$

where $W_O \in \mathbb{R}^{L_t\times d_v}$ is the weight matrix of the alignment layer and $b \in \mathbb{R}^{L_t\times N_{\text{gpt2}}}$ is the bias term.

In another branch, for real-time measurement data m_i, an FFN is directly used to align it to the N_{gpt2} dimension, as expressed below:

$$h_{m_i} = \text{FFN}(m_i) = \text{ReLU}(\text{ReLU}(m_i W_1 + b_1)W_2 + b_2) \tag{18}$$

where $W_1 \in \mathbb{R}^{N_v\times N_{\text{gpt2}}}$ and $W_2 \in \mathbb{R}^{N_{\text{gpt2}}\times N_{\text{gpt2}}}$ are the weight matrices of the FFN, respectively, $b_1 \in \mathbb{R}^{L_d\times N_{\text{gpt2}}}$ and $b_2 \in \mathbb{R}^{L_d\times N_{\text{gpt2}}}$ are the bias terms.

In the following case study analysis of this research, extensive experiments are conducted to validate the proposed data processing paradigm. The results demonstrate the advantages of the proposed method in terms of both computational efficiency and accuracy.

IV. Experiments

A. Data Generation

Two distribution systems of different scales are introduced in this study to validate the proposed method: *1) IEEE 33-Node Distribution Network:* This system consists of 33 nodes and 37 lines, where Node 1 is directly connected to the external grid. No distributed energy resources are present at any nodes in the network. *2) 1-MV-urban–0-sw Medium-Voltage Distribution Network from the Simbench Library [27]:* This network comprises 144 nodes and 147 lines, with 134 nodes hosting distributed energy resources, including wind, solar, and hydropower.

The load and generation data for all nodes are time series data derived from real-world scenarios, covering an entire year with a resolution of 15 minutes. To obtain the data used during model training, we perform power flow simulations using the pandapower library [32]. The dataset is then divided into training, validation, and test sets in an 80%/10%/10% ratio. For each estimation instance, the text prompt data is tokenized to a fixed length of 128 tokens, with padding tokens applied where necessary to maintain consistent length. The time series data for each instance encompasses a 4-hour window, comprising 16 data points sampled at 15-minute intervals and the patch size for the time series data is set to 4. All computational procedures are executed on Nvidia RTX 4090 GPUs.

B. Model Configuration

To validate the effectiveness of the proposed method, the DSSE approach is benchmarked against the traditional Weighted Least Squares (WLS) method and several recently proposed data-driven DSSE models. The estimation accuracy is evaluated under identical input data conditions, including comparisons with MLP [12], LSTM [13], GNN [14], and PIGNN models [16]. All the comparative approach are implemented according to the original papers, with the same hyperparameters and training configurations. The LSTM model is applied with time series data, while the other models are fed with the real-time measurement data. In particular, the comparative experiments also includes a state estimation

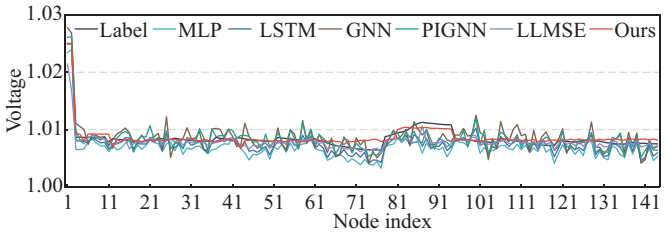


Fig. 2. Comparative voltage results of different approaches in 1-MV-urban–0-sw system.

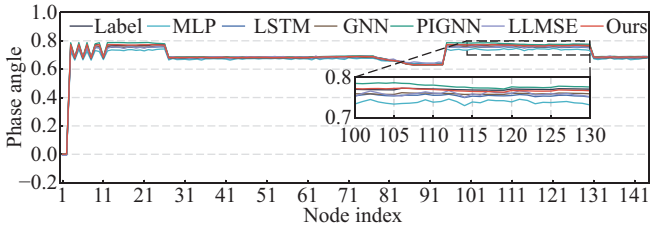


Fig. 3. Comparative phase angle results of different approaches in 1-MV-urban–0-sw system.

model based on GPT-2, further validating the effectiveness of the proposed cross-modal fusion mechanism [26].

For the LLM, we employ the smallest GPT-2 variant, comprising 117 million trainable parameters and 12 transformer decoder layers. The model utilizes a vocabulary of 50,257 tokens, where each token is mapped to a unique index and subsequently transformed into a 768-dimensional word embedding vector. The model implementation is accessed through the Hugging Face Transformers library using the keywords: *openai-community/gpt2*.

GPT-2 was pre-trained on the WebText dataset, which consists of approximately 8 million documents (about 40 GB of text) collected from web pages linked on Reddit. Only pages with at least three karma points were included, reflecting community endorsement, while Wikipedia content was intentionally excluded to avoid overlap with common evaluation datasets [29]. This large-scale and diverse corpus enables GPT-2 to acquire robust language understanding and reasoning capabilities, supporting its effective adaptation to downstream tasks through fine-tuning.

All models employ the Adam optimizer with the MSE loss function and a learning rate of 1×10^{-5}. The exponential decay rate is set to 0.9, and the batch size is configured as 64. Separate models are employed for the estimation of voltage and phase angles. The results produced by all models are evaluated using two statistical metrics: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). To ensure that all models achieve optimal performance in the case studies, we conducted a grid search for hyperparameter configuration on each model. The search ranges are presented in Table II.

TABLE II
HYPERPARAMETER CONFIGURATIONS AND GRID SEARCH SPACES FOR DIFFERENT METHODS

Method	Hyperparameter	Search Space
MLP	Hidden size	{128, 256, 512, 1024}
	Number of layers	{3, 5, 7}
LSTM	Hidden size	{128, 256, 512, 1024}
	Number of LSTM layers	{3, 5, 7}
GNN	Hidden size	{128, 256, 512, 1024}
	Number of layers	{3, 5, 7}
PIGNN	Hidden size	{128, 256, 512, 1024}
	Number of layers	{3, 5, 7}
LLMSE	Hidden size of embedding layer	{128, 256, 512, 1024}
	Hidden size of output layer	{128, 256, 512, 1024}
Ours	Hidden size of alignment layer	{128, 256, 512, 1024}
	Hidden size of time series input layer	{128, 256, 512, 1024}
	Hidden size of RT measurement layer	{128, 256, 512, 1024}
	Hidden size of output layer	{128, 256, 512, 1024}

C. Performance Evaluation of the Proposed DSSE Method

We evaluated the effectiveness of our proposed DSSE method in two scenarios. Following [26], consistent measurement configurations were implemented across all methods, incorporating historical operational data of active power (P) and reactive power (Q) from all nodes, along with real-time power measurements from 60% of the lines (assuming measurement devices are installed on only 60% of the lines). Gaussian noise with a standard deviation of 5% was added to the real-time measurement data to simulate measurement errors. For both LLMSE and our proposed method, identical text prompts were utilized as model inputs, designed according to Section III-B.

Table III presents a comprehensive comparison of performance metrics between the benchmark methods and our proposed approach. The results demonstrate several key findings. First, in under-measured scenarios, data-driven methods generally outperform the WLS approach, with neural networks incorporating explicit prior information exhibiting superior estimation performance. Second, textual prompt information substantially enhances DSSE accuracy under limited measurement conditions, as evidenced by the improved results

TABLE III
PERFORMANCE COMPARISON OF DIFFERENT DSSE METHODS

Methods	IEEE 33-Node Distribution Network				1-MV-urban-0-sw			
	Voltage		Phase angle		Voltage		Phase angle	
	MAE (10^{-3})	RMSE (10^{-3})	MAE (10^{-3})	RMSE (10^{-3})	MAE (10^{-3})	RMSE (10^{-3})	MAE (10^{-3})	RMSE (10^{-3})
WLS	24.2	24.4	18.8	28.5	10.5	18.7	39.6	42.1
MLP [12]	17.0	17.4	8.16	10.1	8.43	9.56	20.5	23.0
LSTM [13]	16.2	17.1	7.98	8.72	5.89	6.45	15.8	16.9
GNN [14]	15.1	16.9	7.87	8.56	4.29	5.13	14.9	22.6
PIGNN [16]	14.2	16.0	7.06	7.62	4.10	4.32	14.7	20.5
LLMSE [26]	9.80	10.6	4.45	4.88	2.02	2.71	11.5	12.3
Ours	**7.78**	**8.22**	**3.79**	**4.20**	**1.50**	**2.40**	**8.00**	**9.46**

The best results are highlighted in **bold**.

achieved by both models utilizing textual information across both scenarios. Finally, while both methods are based on the pre-trained GPT-2 model, our proposed approach demonstrates superior performance compared to LLMSE, highlighting how proper-designed multimodal information processing channels can more effectively utilize limited data to achieve more accurate estimations. Specifically, compared to the second-best performing method, our model achieved reductions in MAE, decreasing to 17.7% and 25.1% of the original values in the two scenarios, respectively.

D. Performance Analysis of Different RTMeasurements Ratios

To thoroughly evaluate the estimation capability of our proposed method in under-measured scenarios, we varied the proportion of real-time measurement devices installed on the lines to [30%, 40%, 50%, 60%, 70%] and provided the same dataset to all the aforementioned models for comparison. Table IV quantifies the state estimation results under varying real-time measurement conditions using the MAE metric (10^{-3}).

TABLE IV
MAE COMPARISON OF DIFFERENT RTMEASUREMENTS RATIO ON DIFFERENT METHODS

Methods	RTMeasurements ratio (%)				
	30	40	50	60	70
MLP	15.4	12.6	9.96	8.43	8.01
LSTM	8.58	7.39	6.78	5.89	5.45
GNN	7.05	6.23	5.38	4.29	4.03
PIGNN	6.33	5.57	4.97	4.10	3.85
LLMSE	2.33	2.21	2.14	2.02	1.98
Ours	1.83	1.66	1.59	1.50	1.42

The results indicate that estimation methods utilizing text prompt inputs significantly outperform other methods, demonstrating that text inputs, when mapped through the LLM, exhibit a stronger representational capacity for the original data. Within our proposed method, even with only 30% of the real-time measurement data, the state estimation performance still surpasses that of LLMSE. This result highlights that, despite both methods leveraging the GPT-2 model for DSSE, the multimodal data processing approach in our method achieves superior feature fusion effectiveness.

In addition, to further validate the effectiveness of our proposed method under varying proportions of available real-time measurement data, we conducted detailed state estimation accuracy assessments with measurement device deployment ratios ranging from 0% to 70%. The results were quantified using the MAE metric. The experimental protocol employed the 1-MV-urban-0-sw system, wherein we performed ten independent random samplings for each measurement device deployment ratio, thereby generating sample sets with identical measurement proportions but distinct measurement location distributions. We carefully ensured diversity and randomness in sampling to evaluate the adaptability of our method under different measurement conditions. Fig. 4 presents box plots illustrating error distributions, clearly demonstrating that our method maintains stable and accurate performance across all measurement levels.

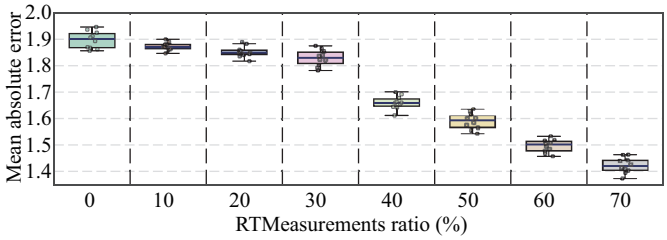


Fig. 4. Performance analysis of the proposed DSSE method under different RTMeasurement installation ratios in 1-MV-urban-0-sw system.

E. Ablation Study of Different Components

To thoroughly validate the effectiveness of each component in our proposed model, we conducted detailed ablation experiments using voltage estimation scenarios on the 1-MV-urban-0-sw distribution network. The evaluation results in Table V clearly demonstrate that removing any component degrades model performance. Specifically, eliminating textual prompt information (**-TextInput**) had the most profound impact, increasing the MAE by 40%. This highlights the positive contribution of textual semantic information, as expressed through the GPT model, to the DSSE task. Removing real-time measurement data (**-RTMeasurement**) or time series data (**-TimeSeriesInput**) increased MAE by 27% and 6% respectively, indicating that real-time information—as a direct reflection of the system's current state—plays a crucial role, while historical data provides valuable but relatively less contribution to the estimation task.

TABLE V
ABLATION RESULT OF DIFFERENT COMPONENTS

Method	MAE (10^{-3})	RMSE (10^{-3})	Degradation rate (%)
Original model	**1.50**	**2.40**	–
-TextInput	2.10	3.00	+140%
-TimeSeriesInput	1.60	2.74	+106%
-RTMeasurement	1.90	2.80	+127%
-GPT-2+Linear	1.80	2.77	+120%

The best results are highlighted in **bold**.

Furthermore, we implemented an ablation strategy that completely replaced GPT-2 layers with linear layers (**-GPT-2+Linear**), resulting in a 20% increase in MAE. This further confirms the advantages of incorporating large language models into distribution system state estimation tasks and validates the rationality and effectiveness of our proposed DSSE architecture.

F. Performance Evaluation of the GPT-2 Model

To validate the effectiveness and versatility of the GPT-2 component in our proposed method, we designed it as a plug-and-play state estimation module specifically for processing textual descriptions, which can be integrated with MLP, LSTM, GNN, and PIGNN architectures. We evaluated the capabilities of different model configurations using voltage estimation experiments on the 1-MV-urban-0-sw network, with MAE as the performance metric. The results in Table VI demonstrate that after incorporating the GPT-2 module, all

TABLE VI
PERFORMANCE COMPARISON WITH AND WITHOUT GPT-2 COMPONENT

Methods	MAE (10^{-3})	RMSE (10^{-3})	Average improvement rate (%)
MLP	8.43	9.56	
GPT-2 + MLP	**1.65**	**2.81**	**+425%**
LSTM	5.89	6.45	
GPT-2 + LSTM	**1.94**	**3.11**	**+256%**
GNN	4.29	5.12	
GPT-2 + GNN	**1.62**	**2.79**	**+224%**
PIGNN	4.10	4.32	
GPT-2 + PIGNN	**1.62**	**2.77**	**+205%**

The best results are highlighted in **bold**.

three baseline methods achieved performance levels comparable to our proposed approach. However, their relatively lower estimation accuracy compared to our complete method underscores the crucial role of temporal sequence data in the estimation task.

G. Performance Evaluation of the Pre-trained Parameters

One of the most significant reasons for using LLMs lies in their pre-trained parameters, which are derived from self-supervised training over massive amounts of contextual data. In our DSSE task, we fine-tune these pre-trained parameters, aiming to exploit their advantages—not only to accelerate model convergence but also to improve estimation accuracy under identical input conditions. To validate the effectiveness of the pre-trained parameters, we conducted a controlled experiment by resetting the pre-trained parameters of the GPT-2 model and comparing the loss values of the models. It is important to note that, to ensure the validity of the controlled experiment, only the parameters designated for training in the GPT-2 model are reset, while the frozen parameters remained unchanged. The loss reduction trends for models with and without pre-trained parameters are shown in Fig. 5.

TABLE VII
PERFORMANCE COMPARISON WITH AND WITHOUT PRE-TRAINED PARAMETERS

Methods	IEEE 33-Node DS			1-MV-urban–0-sw		
	Conv. Step	Time (minute)	MAE (10^{-3})	Conv. Step	Time (minute)	MAE (10^{-3})
Without pre-train	122	22.4	8.32	170	26.7	1.94
With pre-train	32	10.7	7.78	50	13.2	1.50

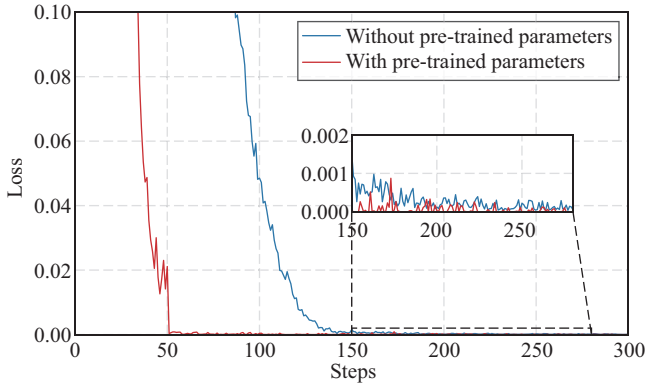


Fig. 5. Loss trends of models with and without pre-trained parameters. The zoomed-in section highlights the convergence level of the models.

The experimental results demonstrate that in both distribution network scenarios, the fine-tuning process based on pre-trained parameters achieves convergence within 32–50 steps. In contrast, models with reset parameters require 122–170 steps to reach comparable performance levels. Furthermore, under identical evaluation criteria, models with reset parameters fail to achieve the same level of convergence as their pre-trained counterparts, indicating the significant advantages of leveraging pre-trained parameters in model optimization.

V. CONCLUSION

This paper proposes a novel framework that integrates large language models with multimodal data fusion to enhance distribution system state estimation under measurement-limited conditions. The framework contributes three key innovations: 1) utilization of LLMs' pattern recognition capabilities for voltage magnitude and phase angle estimation in data-scarce environments, 2) development of a prompt engineering methodology that transforms statistical features from historical data into contextual natural language representations to enhance temporal feature extraction, and 3) implementation of a channel-independent multimodal fusion architecture that structurally aligns time-series data, real-time measurements, and textual prompts while preserving modality-specific characteristics. Experimental validation on IEEE 33-bus and Simbench 144-bus systems demonstrates the effectiveness of the proposed method. Under limited measurement scenarios, the framework achieves 17.7%–25.1% reduction in MAE and 17.3%–18.2% reduction in RMSE compared to state-of-the-art data-driven baselines.

The findings indicate that pre-trained large language model can effectively augment traditional power system analytics when measurement resources are constrained. The framework's capability to extract meaningful patterns from limited data demonstrates its potential for deployment in sparsely instrumented distribution grids. Future research include extending the framework to larger-scale networks, investigating real-time implementation strategies, and exploring uncertainty quantification mechanisms to enhance the reliability of LLM-based state estimation in practical deployment scenarios.

REFERENCES

[1] Q. H. Wu, A. Bose, C. Singh, J. H. Chow, G. Mu, Y. Z. Sun, Z. X. Liu, Z. G. Li, and Y. Liu, "Control and stability of large-scale power system with highly distributed renewable energy generation: viewpoints from six aspects," *CSEE Journal of Power and Energy Systems*, vol. 9, no. 1, pp. 8–14, Jan. 2023.

[2] E. M. Lourenço and J. B. A. London Jr, *Power Distribution System State Estimation*, IET, 2022.

[3] S. Dalhues, Y. Zhou, O. Pohl, F. Rewald, F. Erlemeyer, D. Schmid, J. Zwartscholten, Z. Hagemann, C. Wagner, D. M. Gonzalez, H. Liu, M. M. Zhang, J. Y. Liu, C. Rehtanz, Y. Li, and Y. J. Cao, "Research and practice of flexibility in distribution systems: a review," *CSEE Journal of Power and Energy Systems*, vol. 5, no. 3, pp. 285–294, Sep. 2019.

[4] K. Dehghanpour, Z. Y. Wang, J. H. Wang, Y. X. Yuan, and F. K. Bu, "A survey on state estimation techniques and challenges in smart distribution systems," *IEEE Transactions on Smart Grid*, vol. 10, no. 2, pp. 2312–2322, Mar. 2019.

[5] R. Wang, H. R. Ji, P. Li, H. Yu, J. L. Zhao, L. Zhao, Y. Zhou, J. Z. Wu, L. Q. Bai, J. Y. Yan, and C. S. Wang, "Multi-resource dynamic coordinated planning of flexible distribution network," *Nature Communications*, vol. 15, no. 1, pp. 4576, 2024.

[6] D. C. Hou, Y. H. Sun, L. C. Zhang, and S. Wang, "Robust forecasting-aided state estimation considering uncertainty in distribution system," *CSEE Journal of Power and Energy Systems*, vol. 10, no. 4, pp. 1632–1641, Jul. 2024.
[7] A. Abur and A. G. Exposito, *Power System State Estimation: Theory and Implementation*, Boca Raton: CRC Press, 2004.
[8] K. R. Mestav, J. Luengo-Rozas, and L. Tong, "State estimation for unobservable distribution systems via deep neural networks," in *Proceedings of the 2018 IEEE Power & Energy Society General Meeting (PESGM)*, 2018, pp. 1–5.
[9] X. B. Jin, R. J. R. Jeremiah, T. L. Su, Y. T. Bai, and J. L. Kong, "The new trend of state estimation: from model-driven to hybrid-driven methods," *Sensors*, vol. 21, no. 6, pp. 2085, Mar. 2021.
[10] Y. Guo, Y. F. Xu, H. B. Sun, and B. M. Zhang, "Multi-time interval forecasting-aided state estimation incorporating phasor measurements for power systems with renewable energy sources," *CSEE Journal of Power and Energy Systems*, vol. 11, no. 1, pp. 115–123, Jan. 2025.
[11] D. X. Zhang, X. Q. Han, and C. Y. Deng, "Review on the research and practice of deep learning and reinforcement learning in smart grids," *CSEE Journal of Power and Energy Systems*, vol. 4, no. 3, pp. 362–370, Sep. 2018.
[12] B. Azimian, R. S. Biswas, S. Moshtagh, A. Pal, L. Tong, and G. Dasarathy, "State and topology estimation for unobservable distribution systems using deep neural networks," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 9003514, Apr. 2022.
[13] Z. Y. Cao, Y. B. Wang, C. C. Chu, and R. Gadh, "Scalable distribution systems state estimation using long short-term memory networks as surrogates," *IEEE Access*, vol. 8, pp. 23359–23368, Jan. 2020.
[14] R. Madbhavi, B. Natarajan, and B. Srinivasan, "Graph neural network-based distribution system state estimators," *IEEE Transactions on Industrial Informatics*, vol. 19, no. 12, pp. 11630–11639, Dec. 2023.
[15] O. Kundacina, M. Cosovic, and D. Vukobratovic, "State estimation in electric power systems leveraging graph neural networks," in *Proceedings of the 2022 17th International Conference on Probabilistic Methods Applied to Power Systems (PMAPS)*, 2022, pp. 1–6.
[16] Q. H. Ngo, B. L. H. Nguyen, T. V. Vu, J. H. Zhang, and T. Ngo, "Physics-informed graphical neural network for power system state estimation," *Applied Energy*, vol. 358, pp. 122602, Mar. 2024.
[17] B. Habib, E. Isufi, W. Van Breda, A. Jongepier, and J. L. Cremer, "Deep statistical solver for distribution system state estimation," *IEEE Transactions on Power Systems*, vol. 39, no. 2, pp. 4039–4050, Mar. 2024.
[18] L. Wang, Q. Zhou, and S. S. Jin, "Physics-guided deep learning for power system state estimation," *Journal of Modern Power Systems and Clean Energy*, vol. 8, no. 4, pp. 607–615, Jul. 2020.
[19] C. L. Xia, X. L. Wang, F. Lv, X. Hao, and Y. F. Shi, "ViT-CoMer: vision transformer with convolutional multi-scale feature interaction for dense predictions," in *Proceedings of the 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 5493–5502.
[20] S. Majumder, L. Dong, F. Doudi, Y. T. Cai, C. Tian, D. Kalathil, K. Ding, A. A. Thatte, N. Li, and L. Xie, "Exploring the capabilities and limitations of large language models in the electric energy sector," *Joule*, vol. 8, no. 6, pp. 1544–1549, Jun. 2024.
[21] M. X. Yuan, X. R. Shi, N. Wang, Y. Y. Wang, and X. X. Wei, "Improving RGB-infrared object detection with cascade alignment-guided transformer," *Information Fusion*, vol. 105, pp. 102246, May 2024.
[22] X. Y. Zhou, H. Zhao, Y. H. Cheng, Y. J. Cao, G. Q. Liang, G. L. Liu, W. X. Liu, Y. Xu, and J. H. Zhao, "ElecBench: a power dispatch evaluation benchmark for large language models," arXiv preprint arXiv: 2407.05365, 2024.
[23] H. F. Wang, J. W. Li, H. Wu, E. Hovy, and Y. Sun, "Pre-trained language models and their applications," *Engineering*, vol. 25, pp. 51–65, Jun. 2023.
[24] M. Y. Gao, S. Y. Zhou, W. Gu, Z. Wu, H. Q. Liu, A. H. Zhou, and X. L. Wang, "MMGPT4LF: leveraging an optimized pre-trained GPT-2 model with multi-modal cross-attention for load forecasting," *Applied Energy*, vol. 392, pp. 125965, Aug. 2025.
[25] H. Xue and F. D. Salim, "PromptCast: a new prompt-based learning paradigm for time series forecasting," *IEEE Transactions on Knowledge and Data Engineering*, vol. 36, no. 11, pp. 6851–6864, Nov. 2024.
[26] R. Y. Xie, X. Yin, C. J. Li, G. Chen, N. Liu, B. Zhao, and Z. Y. Dong, "Large language model-aided edge learning in distribution system state estimation," *IEEE Internet of Things Journal*, vol. 12, no. 14, pp. 25966–25976, Jul. 2025.
[27] S. Meinecke, D. Sarajlić, S. R. Drauz, A. Klettke, L. P. Lauven, C. Rehtanz, A. Moser, and M. Braun, "SimBench—a benchmark dataset of electric power systems to compare innovative solutions based on power flow analysis," *Energies*, vol. 13, no. 12, pp. 3290, Jun. 2020.
[28] T. Zhou, P. S. Niu, X. Wang, L. Sun, and R. Jin, "One fits all: power general time series analysis by pretrained LM," in *Proceedings of the 37th International Conference on Neural Information Processing Systems*, 2023, pp. 1877.
[29] A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever, "Language models are unsupervised multitask learners," *OpenAI Blog*, vol. 1, no. 8, pp. 9, 2019.
[30] M. Jin, S. Y. Wang, L. T. Ma, Z. X. Chu, J. Y. Zhang, X. M. Shi, P. Y. Chen, Y. X. Liang, Y. F. Li, S. R. Pan, and Q. S. Wen, "Time-LLM: time series forecasting by reprogramming large language models," in *Proceedings of the 12th International Conference on Learning Representations*, 2024.
[31] Z. H. Zhou and R. Yu, "Can LLMs understand time series anomalies?," in *Proceedings of the 13th International Conference on Learning Representations, ICLR*, 2025.
[32] L. Thurner, A. Scheidler, F. Schäfer, J. H. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, "Pandapower—an open-source python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, Nov. 2018.



Mingyang Gao received the B.S. degree in Electrical Engineering from Southeast University, Nanjing, China, in 2019. He is currently pursuing the Ph.D. degree in Electrical Engineering at Southeast University, Nanjing, China. His research interests include time-series forecasting and applications of pre-trained LLMs in power systems.



Suyang Zhou received the B.Eng. degree in Electrical and Electronic Engineering from Huazhong University of Science and Technology, Wuhan, China in 2009. He received his Ph.D. degree in Electrical and Electronic Engineering at University of Birmingham, Birmingham, U.K. in 2015. He is currently an Associate Professor with the School of Electrical Engineering, Southeast University. His research interests include integrated energy system, big data and demand-side management.



Wei Gu received his B.S. and Ph.D. degrees in Electrical Engineering from Southeast University, China, in 2001 and 2006, respectively. From 2009 to 2010, he was a Visiting Scholar in the Department of Electrical Engineering, Arizona State University. He is currently a Professor at the School of Electrical Engineering, Southeast University. He is the Director of the Institute of Distributed Generations and Active Distribution Networks. His research interests include distributed generations and microgrids, integrated energy systems.



Jili Fan received his B.S. and M.S. degree in Computer Science and Technology in 2021 and 2024 from the Nanjing University of Information Science and Technology, Nanjing, China. He is currently working toward the Ph.D. degree in Electrical Engineering at Southeast University, Nanjing, China. His research interests include data analytics in multiple energy systems and smart grids.



Aobo Guan received the B.S. degree in Electrical Engineering from Southeast University, Nanjing, China, in 2021, where he is currently working toward the Ph.D. degree in Electrical Engineering. His research interests include the modeling, simulation, and operation of electric-hydrogen and electric-heat energy system.



Hong Zhu received the B.S. degree in Computer Science from Southeast University, Nanjing, China, in 1994, and the M.S. degree in Electric Power Systems and Automation from North China Electric Power University, Beijing, China, in 2015. She is currently a Senior Engineer with the State Grid Nanjing Power Supply Company, Nanjing, China. She has received a gold medal at the Geneva Invention Exhibition and two China Patent Awards related to grid digitalization.



Lei Wei received his Ph.D. degree in Energy and Electrical Engineering in Hohai University, Nanjing, China. He is currently the Deputy General Manager of the Information and Communication Branch, State Grid Jiangsu Electric Power Co., Ltd., and a Professor-level Senior Engineer. His research interests focus on power information and communication technology.



Zijian Hu received his B.S. and M.S. degrees in Electrical Engineering from Southeast University, China, in 2018 and 2021, respectively. He is currently an Engineer with the State Grid Nanjing Power Supply Company, Nanjing, China. His research interests include integrated energy system and deep reinforcement learning based techniques.